CSE563 — Unit I Numerical Worksheet

Basic Mathematics in Computer Science

40 questions • CO1 • Solutions Booklet

Instructor Note

Full worked solutions for all 40 questions. Each solution shows the complete method, not just the final answer, so it can be used directly as a model answer on the board. Question numbering matches the student worksheet exactly.

A. Number Theory (Q1–Q10)

Q1. Find the quotient q and remainder r when $a = -73$ is divided by $d = 8$, so that $a = dq + r$ with $0 \le r < d$. State what `-73 % 8` returns in C.

Solution. $-73/8 = -9.125$, so floor down: $q = -10$. Then $r = -73 - 8(-10) = -73 + 80 = 7$. $q = -10$, $r = 7$; check $0 \le 7 < 8$ ✓. In C the quotient truncates to -9, giving $-73 - 8(-9) = -1$, so `-73 % 8` $= -1$.

Q2. Compute $\gcd(2166, 6099)$ using the Euclidean algorithm, showing every division step.

Solution.

$$\begin{aligned} 6099 &= 2(2166) + 1767 \\ 2166 &= 1(1767) + 399 \\ 1767 &= 4(399) + 171 \\ 399 &= 2(171) + 57 \\ 171 &= 3(57) + 0 \end{aligned}$$

$\gcd = 57$. Check: $57 \times 38 = 2166$ ✓, $57 \times 107 = 6099$ ✓.

Q3. Find $\gcd(414, 662)$ and $\operatorname{lcm}(414, 662)$.

Solution. $662 = 1(414) + 248$; $414 = 1(248) + 166$; $248 = 1(166) + 82$; $166 = 2(82) + 2$; $82 = 41(2) + 0$. $\gcd = 2$. $\operatorname{lcm} = \dfrac{414 \times 662}{2} = \dfrac{274068}{2} = \mathbf{137034}$.

Q4. Use the extended Euclidean algorithm to find integers x, y with $138x + 82y = \gcd(138, 82)$. Verify your answer.

Solution. $138 = 1(82) + 56$; $82 = 1(56) + 26$; $56 = 2(26) + 4$; $26 = 6(4) + 2$; $4 = 2(2) + 0$. So $\gcd = 2$.

$$\begin{aligned} 2 &= 26 - 6(4) \\ &= 26 - 6(56 - 2(26)) = 13(26) - 6(56) \\ &= 13(82 - 56) - 6(56) = 13(82) - 19(56) \\ &= 13(82) - 19(138 - 82) = 32(82) - 19(138). \end{aligned}$$

$x = -19$, $y = 32$. Check: $-19(138) + 32(82) = -2622 + 2624 = 2$ ✓.

Q5. Find the multiplicative inverse of 9 modulo 31.

Solution. $31 = 3(9) + 4$; $9 = 2(4) + 1$. So gcd $= 1$ and the inverse exists. $1 = 9 - 2(4) = 9 - 2(31 - 3(9)) = 7(9) - 2(31)$. Hence $9^{-1} \equiv \mathbf{7} \pmod{31}$. Check: $9 \times 7 = 63 = 2(31) + 1$ ✓.

Q6. Determine whether 12^{-1} mod 18 exists. Justify your answer.

Solution. $\gcd(12, 18) = 6 \neq 1$, so **the inverse does not exist**. Reason: if $12x \equiv 1 \pmod{18}$ then $12x - 18k = 1$ for some integer k, but the left side is always a multiple of $\gcd(12, 18) = 6$, and $6 \nmid 1$. Contradiction.

Q7. Compute 5^{38} mod 13 using Fermat's little theorem, and confirm by square-and-multiply.

Solution. 13 is prime and $13 \nmid 5$, so $5^{12} \equiv 1$. $38 = 12(3) + 2$, hence $5^{38} \equiv 5^2 = 25 \equiv \mathbf{12} \pmod{13}$.
Square-and-multiply check: $38 = 100110_2 = 32 + 4 + 2$. $5^2 \equiv 12$; $5^4 \equiv 12^2 = 144 \equiv 1$; $5^8 \equiv 1$; $5^{16} \equiv 1$; $5^{32} \equiv 1$. So $5^{38} \equiv 1 \cdot 1 \cdot 12 = 12$ ✓.

Q8. Compute $\phi(1000)$ and $\phi(97)$.

Solution. $1000 = 2^3 \cdot 5^3$. $\phi(1000) = 1000\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{5}\right) = 1000 \cdot \frac{1}{2} \cdot \frac{4}{5} = \mathbf{400}$. 97 is prime, so $\phi(97) = \mathbf{96}$.

Q9. RSA with $p = 3$, $q = 11$, $e = 7$. Find n, $\phi(n)$ and the private exponent d, then encrypt $M = 5$.

Solution. $n = 33$, $\phi(n) = 2 \times 10 = 20$. $\gcd(7, 20) = 1$ ✓. $d = 7^{-1} \bmod 20$: $20 = 2(7) + 6$; $7 = 1(6) + 1$; $1 = 7 - 6 = 7 - (20 - 2 \cdot 7) = 3(7) - 20$. So $d = \mathbf{3}$. Check $7 \times 3 = 21 \equiv 1 \pmod{20}$ ✓. Encrypt: $C = 5^7 \bmod 33$. $5^2 = 25$; $5^4 \equiv 25^2 = 625 = 18(33) + 31 \equiv 31$; $5^7 = 5^4 \cdot 5^2 \cdot 5 \equiv 31 \times 25 \times 5$. $31 \times 25 = 775 = 23(33) + 16 \equiv 16$; $16 \times 5 = 80 = 2(33) + 14 \equiv \mathbf{14}$.
Decrypt check: $14^3 = 2744 = 83(33) + 5 \equiv 5$ ✓.

Q10. Determine whether 391 is prime. If composite, give its prime factorisation and the number of positive divisors.

Solution. $\sqrt{391} \approx 19.8$, so test primes up to 19: $2, 3, 5, 7, 11, 13, 17, 19$. Odd; digit sum 13 (not div 3); not ending 0/5; $391/7 = 55.86$; $391/11 = 35.5$; $391/13 = 30.1$; $391/17 = \mathbf{23}$ exactly. $391 = 17 \times 23$**, composite.** Both factors prime, each exponent 1, so the divisor count is $(1 + 1)(1 + 1) = \mathbf{4}$ (namely $1, 17, 23, 391$).

B. Set Theory (Q11–Q15)

Q11. $A = \{p, q, r, s\}$. State $|\mathcal{P}(A)|$, the number of proper subsets, and the number of 2-element subsets.

Solution. $|\mathcal{P}(A)| = 2^4 = \mathbf{16}$; proper subsets $= 16 - 1 = \mathbf{15}$; 2-element subsets $= \binom{4}{2} = \mathbf{6}$, namely $\{p, q\}, \{p, r\}, \{p, s\}, \{q, r\}, \{q, s\}, \{r, s\}$.

Q12. $U = \{1, \ldots, 12\}$, $A = \{2, 4, 6, 8, 10, 12\}$, $B = \{3, 6, 9, 12\}$. Compute $A \cap B$, $A \cup B$, $A - B$, $\overline{B}$ and $A \oplus B$.

Solution. $A \cap B = \{6, 12\}$; $A \cup B = \{2, 3, 4, 6, 8, 9, 10, 12\}$; $A - B = \{2, 4, 8, 10\}$; $\overline{B} = \{1, 2, 4, 5, 7, 8, 10, 11\}$; $A \oplus B = \{2, 3, 4, 8, 9, 10\}$. *Check:* $|A \cup B| = 8 = 6 + 4 - 2$ ✓.

Q13. In a class of 80, 50 play cricket, 35 play football and 18 play both. How many play (a) at least one, (b) neither, (c) exactly one?

Solution. **(a)** $|C \cup F| = 50 + 35 - 18 = \mathbf{67}$. **(b)** $80 - 67 = \mathbf{13}$. **(c)** Exactly one $= 67 - 18 = \mathbf{49}$ (cricket only $50 - 18 = 32$, football only $35 - 18 = 17$; $32 + 17 = 49$ ✓).

Q14. Of 150 developers: 85 know Python, 70 know Java, 60 know C++, 35 know Python and Java, 30 know Python and C++, 25 know Java and C++, 15 know all three. Find the number who know at least one, none, and exactly two languages.

Solution.

$$|P \cup J \cup C| = 85 + 70 + 60 - 35 - 30 - 25 + 15 = \mathbf{140}.$$

None: $150 - 140 = \mathbf{10}$. **Exactly two:** $(35 - 15) + (30 - 15) + (25 - 15) = 20 + 15 + 10 = \mathbf{45}$. *Cross-check:* exactly one $= 140 - 45 - 15 = 80$; directly, P only $= 85 - 20 - 15 - 15 = 35$, J only $= 70 - 20 - 10 - 15 = 25$, C only $= 60 - 15 - 10 - 15 = 20$; total 80 ✓.

Q15. $A = \{1, 2\}$, $B = \{x, y, z\}$. Find $|A \times B|$, list three elements of $A \times B$, and state $|\mathcal{P}(A \times B)|$.

Solution. $|A \times B| = 2 \times 3 = \mathbf{6}$. Elements: $(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)$. $|\mathcal{P}(A \times B)| = 2^6 = \mathbf{64}$. Note $(1, x) \neq (x, 1)$ — ordered pairs.

C. Boolean Algebra (Q16–Q21)

Q16. Simplify $F = AB + A\overline{B} + \overline{A}B$ using Boolean laws, naming each law.

Solution.

$$\begin{aligned} F &= A(B + \overline{B}) + \overline{A}B && \text{(distributive)} \\ &= A(1) + \overline{A}B = A + \overline{A}B && \text{(complement, identity)} \\ &= A + B && \text{(redundancy: } x + \overline{x}y = x + y\text{)} \end{aligned}$$

$F = A + B$.

Q17. Simplify $F = \overline{\overline{A} + \overline{B}} + AB\overline{C}$.

Solution. By De Morgan, $\overline{\overline{A} + \overline{B}} = AB$. So

$$F = AB + AB\overline{C} = AB(1 + \overline{C}) = AB(1) = \mathbf{AB}$$

by absorption ($x + xy = x$ with $x = AB$, $y = \overline{C}$).

Q18. Write the canonical SOP and POS for $F(A, B, C) = \sum m(0, 3, 5, 6)$.

Solution. **SOP** (rows where $F = 1$): $m_0 = \overline{A}\,\overline{B}\,\overline{C}$, $m_3 = \overline{A}BC$, $m_5 = A\overline{B}C$, $m_6 = AB\overline{C}$.

$$F = \overline{A}\,\overline{B}\,\overline{C} + \overline{A}BC + A\overline{B}C + AB\overline{C}.$$

POS (rows where $F = 0$, i.e. $1, 2, 4, 7$):

$$F = (A + B + \overline{C})(A + \overline{B} + C)(\overline{A} + B + C)(\overline{A} + \overline{B} + \overline{C}) = \prod M(1, 2, 4, 7).$$

(This is the 3-input odd-parity/XOR complement pattern.)

Q19. Minimise $F(A, B, C) = \sum m(1, 3, 5, 7)$ using a K-map.

Solution. Minterms $1, 3, 5, 7$ are exactly those with $C = 1$.

	BC=00	01	11	10
A=0	0	1	1	0
A=1	0	1	1	0

One group of four covers all the 1s; A and B both vary, $C = 1$ throughout. $F = C$.

Q20. Minimise $F(A, B, C) = \sum m(0, 1, 2, 3, 6, 7)$ using a K-map and write the minimal SOP.

Solution.

	BC=00	01	11	10
A=0	1	1	1	1
A=1	0	0	1	1

Red: the whole $A = 0$ row (four cells, $m_0..m_3$) $\Rightarrow \overline{A}$. **Green:** column pair $BC = 11, 10$ across both rows (m_2, m_3, m_6, m_7) $\Rightarrow B$.

$$\boxed{F = \overline{A} + B}$$

Check m_4 (A=1, B=0, C=0), which must be 0: $\overline{A} = 0$, $B = 0$, so $F = 0$ ✓.

Q21. Minimise $F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$ using a 4-variable K-map.

Solution.

	CD=00	01	11	10
AB=00	1	0	0	1
01	0	1	1	0
11	0	1	1	0
10	1	0	0	1

green: BD

red: $\overline{B}\,\overline{D}$

Green (centre block of four): m_5, m_7, m_{13}, m_{15} — rows $AB = 01, 11$ and columns $CD = 01, 11$. Here $B = 1$ and $D = 1$ throughout, A and C vary $\Rightarrow BD$. **Red (four corners):** m_0, m_2, m_8, m_{10} — $B = 0$ and $D = 0$ throughout $\Rightarrow \overline{B}\,\overline{D}$.

$$\boxed{F = BD + \overline{B}\,\overline{D}} \quad \text{which is } \overline{B \oplus D} \text{ (XNOR).}$$

Check m_1 (B=0, D=1), which must be 0: $BD = 0$, $\overline{B}\,\overline{D} = 1 \cdot 0 = 0$, so $F = 0$ ✓.

D. Logic (Q22–Q27)

Q22. Construct the truth table for $(p \vee q) \to (p \wedge q)$ and classify it.

Solution.

p	q	$p \vee q$	$p \wedge q$	whole
T	T	T	T	**T**
T	F	T	F	**F**
F	T	T	F	**F**
F	F	F	F	**T**

Mixed T and F, so it is a **contingency**. (It is true exactly when p and q agree — i.e. it is equivalent to $p \leftrightarrow q$.)

Q23. Show that $p \to q \equiv \neg q \to \neg p$ using a truth table.

Solution.

p	q	$p \to q$	$\neg q$	$\neg p$	$\neg q \to \neg p$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Columns 3 and 6 are identical, so the two are **logically equivalent** — this is the *contrapositive* law.

Q24. Prove algebraically (no truth table) that $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$.

Solution.

$$\begin{aligned}
\neg(p \vee (\neg p \wedge q)) &\equiv \neg((p \vee \neg p) \wedge (p \vee q)) && \text{(distributive)} \\
&\equiv \neg(\mathrm{T} \wedge (p \vee q)) && \text{(complement)} \\
&\equiv \neg(p \vee q) && \text{(identity)} \\
&\equiv \neg p \wedge \neg q && \text{(De Morgan)} \quad \blacksquare
\end{aligned}$$

Q25. Test the validity of: "If the server is down, users complain. Users are not complaining. Therefore the server is not down." Name the rule used.

Solution. Let d = "server is down", c = "users complain". Premises: $d \to c$, $\neg c$. Conclusion: $\neg d$. This is **modus tollens**, and the argument is **valid**. *Verification:* the only row with $d \to c$ true and c false is $d = \mathrm{F}$, $c = \mathrm{F}$ — in which $\neg d$ is true. There is no row with true premises and a false conclusion.

Q26. Test the validity of: "If it rains, the match is cancelled. The match was cancelled. Therefore it rained."

Solution. Premises: $r \to c$, c. Conclusion: r. **Invalid** — this is the fallacy of *affirming the consequent*. *Counterexample row:* $r = \mathrm{F}$, $c = \mathrm{T}$. Then $r \to c$ is T (vacuously) and c is T, so both premises hold, yet the conclusion r is F. The match could have been cancelled for another reason entirely.

Q27. Negate $\exists x\,\forall y\,(P(x,y)\wedge Q(y))$ and push the negation as far inward as possible.

Solution.

$$\begin{aligned}\neg\exists x\,\forall y\,(P(x,y)\wedge Q(y)) &\equiv \forall x\,\neg\forall y\,(P(x,y)\wedge Q(y))\\ &\equiv \forall x\,\exists y\,\neg(P(x,y)\wedge Q(y))\\ &\equiv \forall x\,\exists y\,(\neg P(x,y)\vee\neg Q(y)).\end{aligned}$$

Each quantifier flips in turn, and De Morgan converts the final $\wedge$ into $\vee$.

E. Calculus (Q28–Q32)

Q28. Differentiate $y=(3x^2+1)^4$.

Solution. Chain rule with outer u^4 and inner $u=3x^2+1$ (so $u'=6x$):

$$\frac{dy}{dx}=4(3x^2+1)^3\cdot 6x=\mathbf{24x(3x^2+1)^3}.$$

Q29. Differentiate $y=x^2\ln x$ and find the value of $x>0$ at which the derivative is zero.

Solution. Product rule: $y'=2x\ln x+x^2\cdot\dfrac{1}{x}=2x\ln x+x=x(2\ln x+1)$. Setting $y'=0$ with $x>0$: $2\ln x+1=0\Rightarrow\ln x=-\frac{1}{2}\Rightarrow x=e^{-1/2}\approx\mathbf{0.6065}$.

Q30. For $f(x,y)=3x^2y+2y^3-5x$, find $\partial f/\partial x$, $\partial f/\partial y$ and ∇f at $(1,2)$.

Solution. $\dfrac{\partial f}{\partial x}=6xy-5$; $\dfrac{\partial f}{\partial y}=3x^2+6y^2$. At $(1,2)$: $\partial f/\partial x=12-5=7$; $\partial f/\partial y=3+24=27$.

$$\nabla f(1,2)=(\mathbf{7},\ \mathbf{27}).$$

Q31. Minimise $L(w)=3w^2-18w+40$ analytically, then perform two gradient-descent steps from $w_0=0$ with $\alpha=0.1$.

Solution. **Analytic:** $L'(w)=6w-18=0\Rightarrow w^*=\mathbf{3}$; $L''=6>0$ so it is a minimum; $L(3)=27-54+40=\mathbf{13}$.
Gradient descent: $L'(0)=-18$, so $w_1=0-0.1(-18)=\mathbf{1.8}$. $L'(1.8)=10.8-18=-7.2$, so $w_2=1.8+0.72=\mathbf{2.52}$. Values: $L(0)=40$, $L(1.8)=9.72-32.4+40=17.32$, $L(2.52)=19.05-45.36+40=13.69$ — decreasing toward 13 ✓.

Q32. Evaluate $\displaystyle\int_0^2(3x^2-4x+1)\,dx$.

Solution. $F(x)=x^3-2x^2+x$.

$$\int_0^2=F(2)-F(0)=(8-8+2)-0=\mathbf{2}.$$

F. Mathematical Induction (Q33–Q36)

Q33. Prove by induction that $\displaystyle\sum_{i=1}^{n} i(i+1) = \frac{n(n+1)(n+2)}{3}$ for all $n \geq 1$.

Solution. **Basis ($n = 1$):** LHS $= 1(2) = 2$; RHS $= \dfrac{1 \cdot 2 \cdot 3}{3} = 2$ ✓.
Hypothesis: $\displaystyle\sum_{i=1}^{k} i(i+1) = \frac{k(k+1)(k+2)}{3}$.
Step:

$$\begin{aligned}\sum_{i=1}^{k+1} i(i+1) &= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2) \\ &= (k+1)(k+2)\left[\frac{k}{3} + 1\right] = (k+1)(k+2) \cdot \frac{k+3}{3} \\ &= \frac{(k+1)\big((k+1)+1\big)\big((k+1)+2\big)}{3},\end{aligned}$$

the formula at $n = k+1$. ■

Q34. Prove by induction that $5 \mid (n^5 - n)$ for all integers $n \geq 0$.

Solution. **Basis ($n = 0$):** $0 = 5(0)$ ✓.
Hypothesis: $k^5 - k = 5t$.
Step: expand $(k+1)^5 = k^5 + 5k^4 + 10k^3 + 10k^2 + 5k + 1$, so

$$\begin{aligned}(k+1)^5 - (k+1) &= k^5 + 5k^4 + 10k^3 + 10k^2 + 5k + 1 - k - 1 \\ &= (k^5 - k) + 5k^4 + 10k^3 + 10k^2 + 5k \\ &= 5t + 5(k^4 + 2k^3 + 2k^2 + k) = 5\big(t + k^4 + 2k^3 + 2k^2 + k\big),\end{aligned}$$

divisible by 5. ■

Q35. Prove by induction that $n! > 2^n$ for all $n \geq 4$.

Solution. **Basis ($n = 4$):** $4! = 24 > 16 = 2^4$ ✓.
Hypothesis: $k! > 2^k$ for some $k \geq 4$.
Step: $(k+1)! = (k+1) \cdot k! > (k+1)2^k$. Since $k \geq 4$ we have $k + 1 \geq 5 > 2$, so

$$(k+1)2^k > 2 \cdot 2^k = 2^{k+1}.$$

Hence $(k+1)! > 2^{k+1}$. ■

Q36. Use strong induction to prove that every integer $n \geq 8$ can be written as a sum of 3s and 5s.

Solution. **Basis:** $8 = 3 + 5$; $9 = 3 + 3 + 3$; $10 = 5 + 5$. (Three base cases are needed because the step reaches back by 3.)
Hypothesis: assume every integer m with $8 \leq m \leq k$ is expressible, for some $k \geq 10$.
Step: consider $k + 1 \geq 11$. Then $(k+1) - 3 = k - 2 \geq 8$, so by the strong hypothesis $k - 2$ is expressible as a sum of 3s and 5s. Adding one more 3 expresses $k + 1$. ■

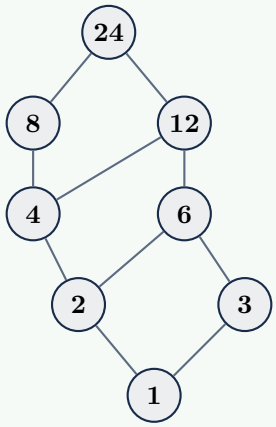
G. Relations and Functions (Q37–Q40)

Q37. On $A = \{1, 2, 3, 4\}$ let $R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (3,4), (4,3)\}$. Determine whether R is an equivalence relation, and if so list its equivalence classes.

Solution. **Reflexive:** $(1,1), (2,2), (3,3), (4,4)$ all present ✓. **Symmetric:** $(1,2)\&(2,1)$, $(3,4)\&(4,3)$ ✓. **Transitive:** check the composable pairs — $(1,2)(2,1) \to (1,1)$ ✓; $(2,1)(1,2) \to (2,2)$ ✓; $(3,4)(4,3) \to (3,3)$ ✓; $(4,3)(3,4) \to (4,4)$ ✓. No pair links $\{1,2\}$ to $\{3,4\}$, so nothing else to check ✓.
R **is an equivalence relation.** Classes: $[1] = [2] = \{1,2\}$ and $[3] = [4] = \{3,4\}$ — two classes partitioning A.

Q38. Draw the Hasse diagram of $(D_{24}, |)$ where $D_{24} = \{1, 2, 3, 4, 6, 8, 12, 24\}$. State the least and greatest elements and find $\text{glb}(8, 12)$ and $\text{lub}(8, 12)$.

Solution.



Least element: 1. **Greatest element:** 24. $\text{glb}(8, 12) = \gcd(8, 12) = \mathbf{4}$; $\text{lub}(8, 12) = \text{lcm}(8, 12) = \mathbf{24}$. 8 and 12 are incomparable, so this is not a total order.

Q39. For $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x + 7$ and $g(x) = x^2$, compute $(f \circ g)(3)$, $(g \circ f)(3)$ and $f^{-1}(x)$.

Solution. $(f \circ g)(3) = f(g(3)) = f(9) = 18 + 7 = \mathbf{25}$. $(g \circ f)(3) = g(f(3)) = g(13) = \mathbf{169}$. They differ — composition is not commutative. f is bijective on $\mathbb{R}$; solving $y = 2x + 7$ gives $f^{-1}(x) = \dfrac{x - 7}{2}$. *Check:* $f^{-1}(f(3)) = f^{-1}(13) = 6/2 = 3$ ✓.

Q40. Evaluate $\lfloor -4.3 \rfloor$, $\lceil -4.3 \rceil$, $\lfloor \log_2 500 \rfloor$. How many bits are needed to represent the decimal number 500, and how many 512-byte disk blocks are needed for a 5000-byte file?

Solution. $\lfloor -4.3 \rfloor = \mathbf{-5}$ (round toward $-\infty$); $\lceil -4.3 \rceil = \mathbf{-4}$. $2^8 = 256 \le 500 < 512 = 2^9$, so $\lfloor \log_2 500 \rfloor = \mathbf{8}$. **Bits for 500:** $\lfloor \log_2 500 \rfloor + 1 = \mathbf{9}$ bits. **Disk blocks:** $\lceil 5000/512 \rceil = \lceil 9.77 \rceil = \mathbf{10}$ blocks.